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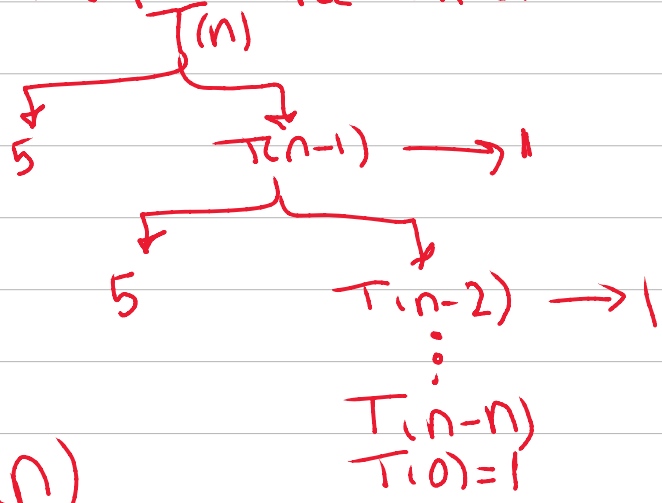
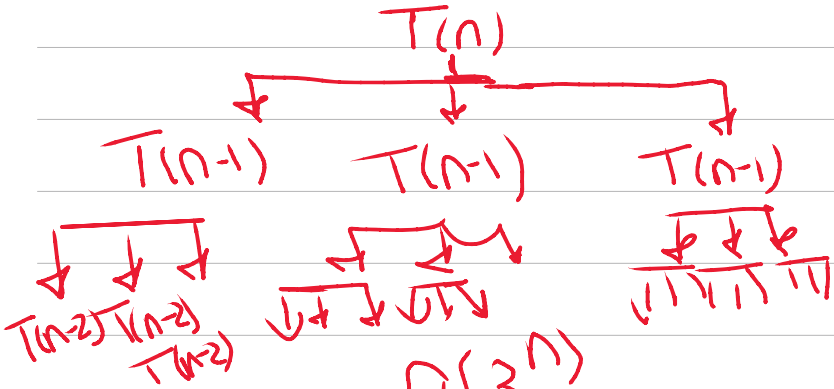
Tree Method Recurrence Sheet



$O(n)$



$O(3^n)$

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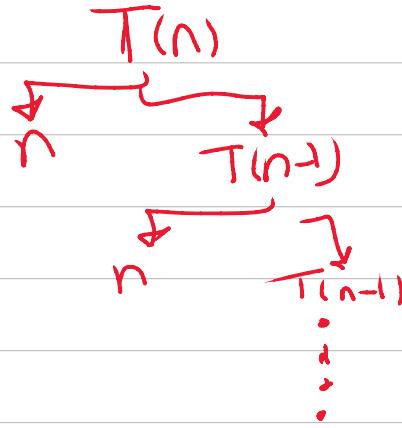
③



$O(n)$

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④ $\log_2 1 = 0$ $\quad n^k \log^p$

$\log_b a < k$ Case III $\quad n^1 \log n^0$

$p = 0$ Case 1 $\quad k = 1 \quad p = 0$

$n^k \log_n p \quad n^1 \log n^0 \quad O(n)$

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⑤ $\log_3 1 = 0$ $\quad n^0 \log n^0$

$k = 0 \quad p = 0$

$\log_b a = k$ Case II $\quad p > -1$

$O(n^k \log n^{p+1})$

$n^0 \log n = \log(n)$

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$T(n)$

n^3 $\quad$ $T(n-1) \rightarrow n^3$

$(n-1)^3$ $\quad$ $T(n-2) \rightarrow n^3$

$(n-2)^3$ $\quad$ $T(n-3)$

$\vdots$

$O(n^4)$

$n * n^3 = O(n^4)$

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7 $\log_b a = 1$ $\quad$ $n^1 \log n^0$

$\log_b a = K$ $\quad$ $K=1$ $\quad$ $P=0$

Case II $\quad$ $P > -1$

$O(n^K \log n^{P+1})$

$n^1 \log n^{0+1} = O(n \log(n))$

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(8) $\log_b a = \log_3 4 \approx 1.2618$

$\log_b a > K$ $\quad n^1 \log n^0$

case I $\quad K=1 \quad P=0$

$O(n^{1.2618})$

(9) $\log_b a = \log_2 2 = 1$ $\quad n^0 \log n^1$

$\log_b a > K$ $\quad K=0 \quad P=1$

Case II

$O(n)$

(10) $\log_b a = 1$ $\quad n^2 \log n^0$

$\log_b a < 2 = K$ $\quad K=2 \quad P=0$

Case III $\quad P \geq 0$ Cas I

$O(n^K \log n^P) \rightarrow O(n^2 \log n)$

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(11) $\log_b a = \log_4 2 = 0.5$

$n^{1/2} \log n^0$

$\log_b a = K$ $\quad K = 1/2, \; P = 0$

Case II

$P > -1$ Case I

$O(n^K \log n^{P+1})$

$O(n^{1/2} \log n^{0+1}) = O(\sqrt{n} \log(n))$

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(2).

(1)

$$T(n) = \begin{cases} 1 & n=1 \\ 2T(n-1) & n>1 \end{cases}$$

$O(n)$

(2)

$$T(n) = \begin{cases} 1 & n=1 \\ T(n-1)+T(n-2) & n>1 \end{cases}$$

$O(n)$

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$$T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) + n & n > 1 \end{cases}$$

using master theorem of dividing function

$\log_b a = 1$ $\quad n^1 \log n^0$

$\log_b a = k$ $\quad k = 1 \quad p = 0$

Case II $\quad p > -1$

Case I in Case II

$O(n^k \log n^{p+1})$

$n^1 \log n^1 \quad O(n \log n)$

(4)

$$T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) + n^2 & n > 1 \end{cases}$$

$\log_b a = 1$ $\quad n^2 \log n^0$

Case III, Case I $\quad k = 2 \quad p = 0$

$O(n^2 \log n^0)$ $\quad O(n^2)$ $\quad p > 1/0$

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(5)

$$T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) + n\sqrt{n} & n > 1 \end{cases}$$

Using Master theorm

$\log_b a = 1$

$n^{1.5} \log n^0$

$K = 1.5$, $P = 0$

Case III, 2

$O(n^{1.5} \log n)$

$O(n\sqrt{n} \log n)$

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Q6

$$T(n) = \begin{cases} 1 & n=1 \\ nT(n-1) + O(n) & n>1 \end{cases}$$

$$T(n) = nT(n-1) * (n-1)T(n-2) * n-2T(n-3)$$

$$O(n!)$$